Applied NLP

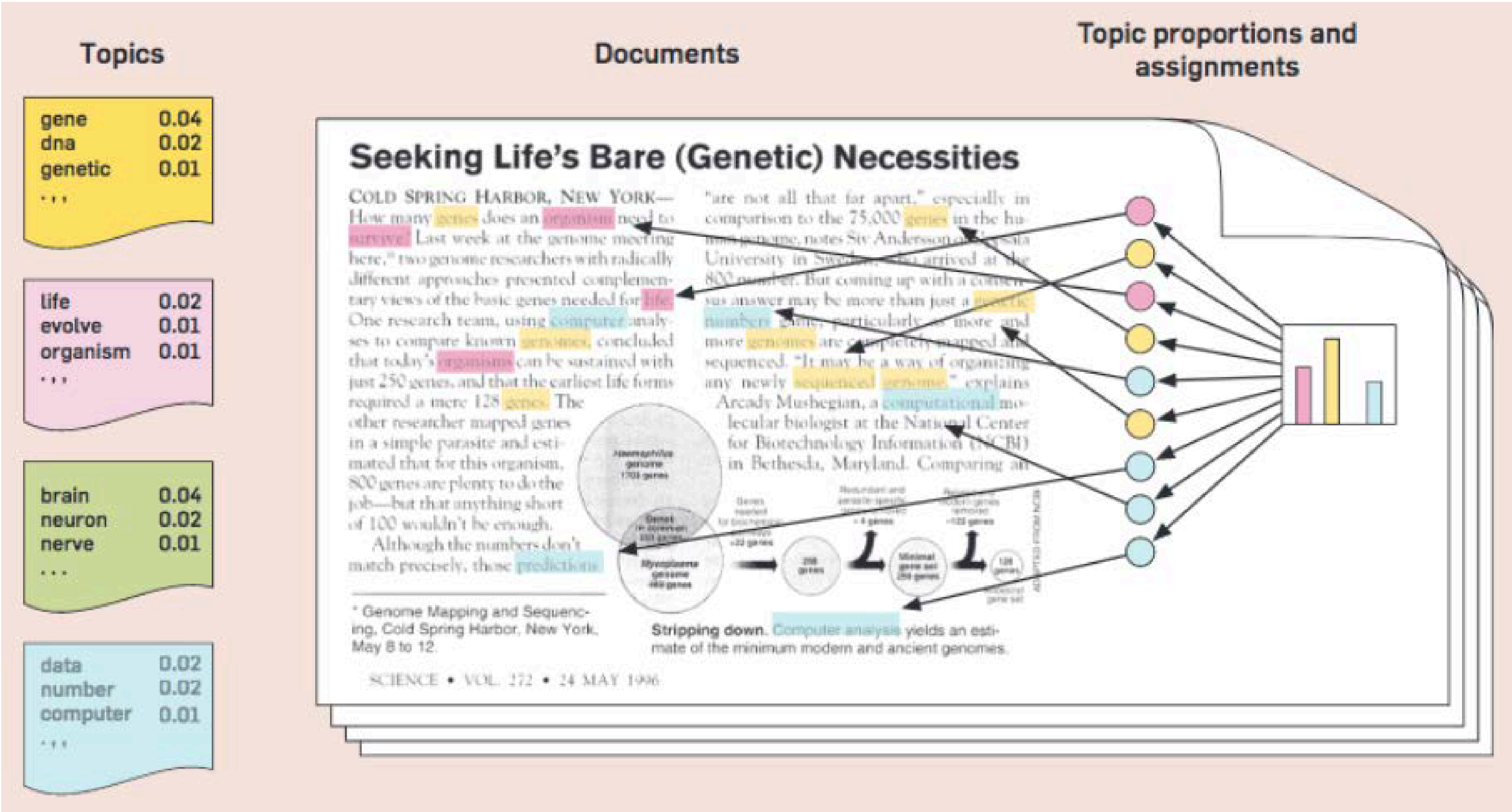
DATA 641

Lecture 13 — Singular value decomposition for topic modeling

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Topic modeling

Topic modeling, a fundamental technique in natural language processing (NLP), helps uncover underlying themes in large collections of text data



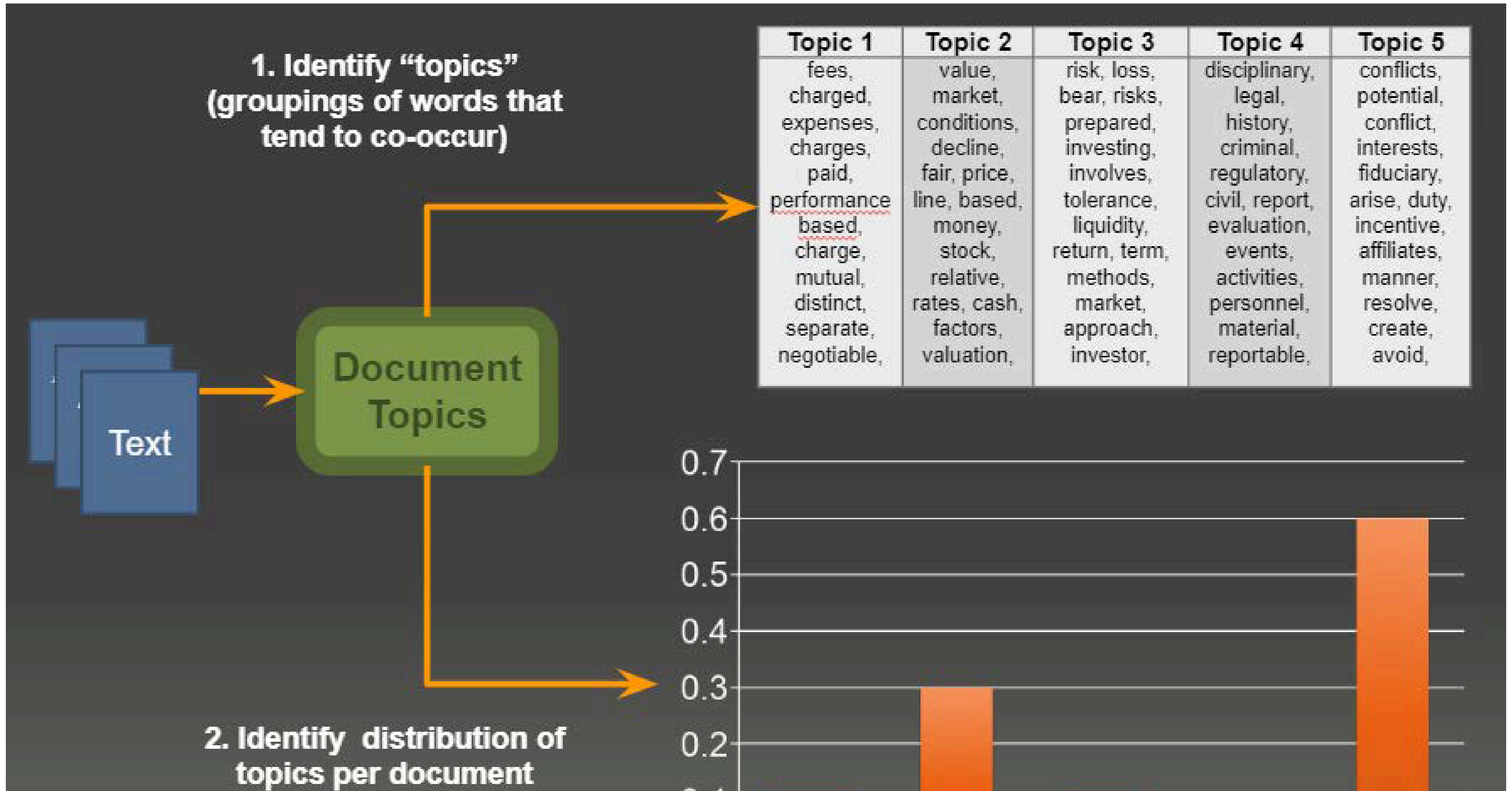
Basic idea behind topic modeling

We know that BoW or other such methods produce matrices!

- Analyzing those matrices will tell us something about the corpus

What happens if we use matrix decomposition algorithms?

- Discover latent trends found in a corpus
- Also known as: "topic modeling"



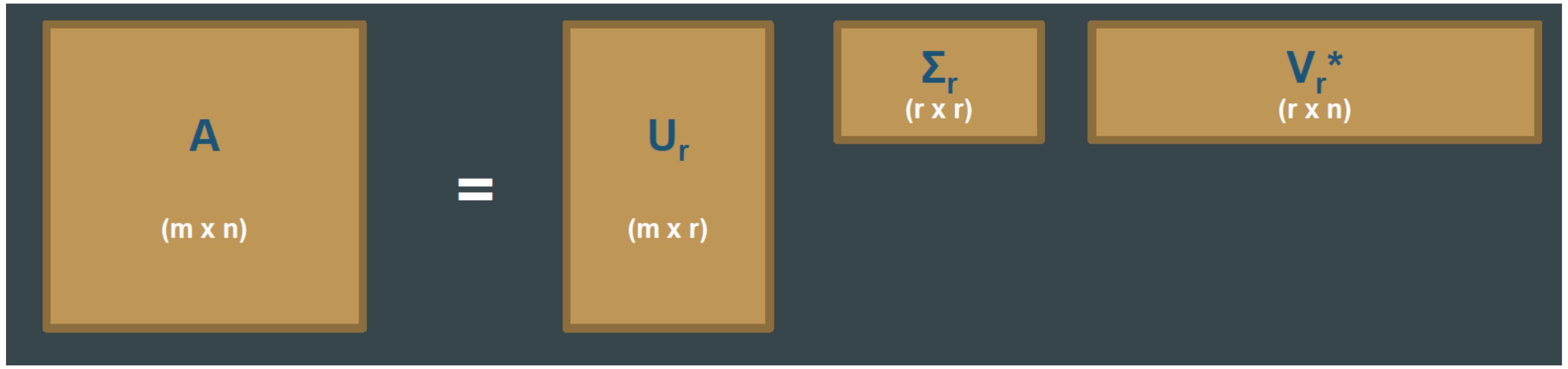
Topic Modeling via Latent Semantic Analysis

Imagine you have a big pile of documents, and you want to understand what they're all about. SVD helps by breaking this problem into three parts:

- $\mathbf{U}_r$: This is like a dictionary or a list of important topics. It assigns words from the documents to these topics. For example, if "school" is a topic, $\mathbf{U}_r$ tells us that it's related to words like "classroom," "teacher," and "student." It's like finding groups of related words.
- $\mathbf{\Sigma}_r$: Think of this as a ranking of topics by importance. Some topics might be super important, while others are less so. $\mathbf{\Sigma}_r$ tells us how crucial each topic is in understanding our documents.
- $\mathbf{V}_r^*$: This part reveals how much of each topic is in each document. It's like a recipe that says, "Document A is 30% about topic 1, 50% about topic 2, and so on." $\mathbf{V}_r^*$ helps us see which topics are present in each document.

Latent Semantic Analysis via SVD

- Use SVD to come up with a rank$-r$ approximation to the matrix $\mathbf{A}$:



- $\mathbf{U}_r$ assigns m terms to r topics. Topic is a group of related terms (e.g., school: {classroom, teacher, student})
- $\mathbf{\Sigma}_r$ quantifies the importance of each topic
- $\mathbf{V}_r^*$ contains topic distribution per each of the n documents

Relationship between SVD and PCA

- PCA can be seen as a specific application of SVD. When you perform PCA on a dataset, you are essentially applying SVD to the covariance matrix of the data.

- PCA involves finding the eigenvalues and eigenvectors of the covariance matrix. These eigenvectors represent the principal components, and the eigenvalues quantify the variance explained by each principal component.

- In SVD, $\mathbf{U}$ contains the left singular vectors, and $\mathbf{\Sigma}$ contains the singular values.

- In PCA, the principal components (eigenvectors of the covariance matrix) can be found in the columns of $\mathbf{U}$ obtained through SVD, and the eigenvalues of the covariance matrix are related to the squared singular values in $\mathbf{\Sigma}$.

- In SVD, you start with a data matrix $\mathbf{A}$ (centered or uncentered) and perform SVD on it. The singular values are obtained as the square roots of the eigenvalues of the matrix $\mathbf{A}^T\mathbf{A}$ (or $\mathbf{A}\mathbf{A}^T$), which is a symmetric positive-semidefinite matrix.

- The singular values σ_i of $\mathbf{A}$ are related to the eigenvalues λ_i of the covariance matrix $\mathbf{C}$ of the centered data as follows

$$\sigma_i = \sqrt{\lambda_i} \tag{1}$$

Advantages of SVD for Topic Modeling

- **Dimensionality Reduction**: SVD can effectively reduce the dimensionality of high-dimensional document-term matrices, such as TF-IDF matrices. This reduction simplifies the data while preserving the most important information related to topics.

- **Noise Reduction**: By focusing on the dominant singular values and corresponding singular vectors (eigenvectors), SVD helps filter out noisy and less informative components. This can improve the quality of topic modeling results by emphasizing the most meaningful patterns.

- **Topic Interpretability**: SVD-based topic modeling often results in topics that are more interpretable and coherent because the topics are derived from linear combinations of terms. This makes it easier to understand and label the discovered topics.

- **Mathematical Robustness**: SVD is a well-established mathematical technique with robust numerical properties. It guarantees the decomposition of a matrix into singular values and vectors, making it a reliable choice for topic modeling.

Disadvantages of SVD for Topic Modeling

- **Lack of Term Independence**: SVD assumes that terms are independent of each other within topics. In reality, words can be correlated, and this assumption may not hold, potentially leading to less accurate topic representations.
- **Loss of Local Context**: SVD focuses on global patterns and relationships between terms and documents but does not capture the local context of words within documents. Contextual information can be important for certain NLP tasks.
- **Sparse Data Handling**: When dealing with sparse matrices, SVD can be computationally expensive and memory-intensive.
- **Difficulty in Handling Streaming Data**: SVD typically requires the entire dataset to be available upfront for decomposition. It is not well-suited for streaming or incremental updates of data.
- **Interpretation of Negative Values**: Some elements in the reduced-dimensional representation obtained through SVD may have negative values, making their interpretation less intuitive.
- **Determining the Number of Topics**: SVD does not inherently provide a method for automatically determining the optimal number of topics. Researchers often need to experiment with different numbers of topics and rely on external evaluation metrics or domain knowledge.